Evaluation of integrals in transport calculations within the DMFT

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I. INTRODUCTION

The behavior of single-particle excitations in a material can be theoretically described using the formalism of Green's functions. We define the spectral function ρ and the Green's function G as

$$
\begin{aligned}
\rho(\omega,\mathbf{k}) &= -\frac{1}{\pi}\mathrm{Im}G(\omega,\mathbf{k}),\\
G(\omega,\mathbf{k}) &= \frac{1}{\omega+\mu-\epsilon_{\mathbf{k}}-\Sigma(\omega,\mathbf{k})},
\end{aligned}
\tag{1}
$$

where $\epsilon_{\mathbf{k}}$ is the energy of an electron with momentum $\mathbf{k}$, ω is frequency, μ is the chemical potential, and $\Sigma(\omega,\mathbf{k})$ is called the self-energy function which describes the effects of interactions. $\Sigma(\omega,\mathbf{k})$ is a complex function such that $\mathrm{Im}\Sigma(\omega,\mathbf{k})<0$ for all arguments, which is a consequence of causality and guarantees that the spectral function $\rho(\omega,\mathbf{k})$ is a positive quantity.

The conductivity of a material is related to the current-current correlations (Kubo's formula). The current, in turn, is proportional to the velocity of electrons $v(\mathbf{k})=d\epsilon_{\mathbf{k}}/dk$ and their distribution function $A(\mathbf{k})$ (the fraction of electrons with momentum $\mathbf{k}$). The correlators are quadratic in $v(\mathbf{k})$ and $A(\mathbf{k})$, and we need to sum over contributions from all $\mathbf{k}$. In some cases, the $\mathbf{k}$ dependence enters only through the dependence of $\epsilon_{\mathbf{k}}$, and the $\mathbf{k}$ sum can be replaced by an integral over electron energy $\epsilon=\epsilon_{\mathbf{k}}$.

In the limit of infinite dimensions, the self-energy becomes purely local, $\Sigma=\Sigma(\omega)$, and there are no vertex corrections. The conductivity can then be evaluated from a simple double integration over electron energy ϵ and frequency ω (bubble formula). This holds for other transport properties as well. For metals, the integrals become difficult to evaluate in the limit of low temperatures because of sharply peaked (almost singular) contributions close to the Fermi level, because for a Fermi-liquid system the self-energy Σ behaves as

$$
\Sigma(\omega)\approx \mathrm{Re}\Sigma(0)+\omega\left(1-\frac{1}{Z}\right)-i\frac{1}{Z^2}[(\pi k_B T)^2+\omega^2],
\tag{2}
$$

thus the broadening ($\mathrm{Im}\Sigma$) becomes small in the $\omega\to 0$, $T\to 0$ limit. Here $0<Z\leq 1$ is the quasiparticle residue, k_B is the Boltzmann constant, and T is the temperature. More generally, such intergrals are difficult to evaluate whenever $\mathrm{Im}\Sigma$ becomes small.

Our goal is to perform the ϵ integration in the bubble formula analytically and reduce the problem to a simpler numerical quadrature over ω only. We need an accurate and precise result in short time. The code needs to be robust with respect to $\Sigma(\omega)$ because the self-energy can be affected with numerical noise and is not necessarily very smooth.

In the following $k_B=\hbar=1$.

II. PROBLEM DEFINITION

We consider the (dimensionless) integrals

$$
I_{mno}=\frac{1}{D}\int_{-\infty}^{\infty}\mathrm{d}\epsilon\,\Phi_m(\epsilon)\int_{-\infty}^{+\infty}\mathrm{d}\omega\left(-\frac{\partial f(\omega)}{\partial\omega}\right)[D\rho(\omega,\epsilon)]^n\left(\frac{\omega}{D}\right)^o,
\tag{3}
$$

where $\Phi_m(\epsilon)$ is one of the (dimensionless) functions

$$
\begin{aligned}
\Phi_1(\epsilon)&=1\times b(\epsilon),\\
\Phi_2(\epsilon)&=\epsilon/D\times b(\epsilon),\\
\Phi_3(\epsilon)&=[1-(\epsilon/D)^2]^{1/2}\times b(\epsilon),\\
\Phi_4(\epsilon)&=(\epsilon/D)[1-(\epsilon/D)^2]^{1/2}\times b(\epsilon),\\
\Phi_5(\epsilon)&=[1-(\epsilon/D)^2]^{3/2}\times b(\epsilon),\\
\Phi_6(\epsilon)&=(\epsilon/D)[1-(\epsilon/D)^2]^{3/2}\times b(\epsilon),\\
\Phi_7(\epsilon)&=\exp[-2(\epsilon/\sigma)^2],\\
\Phi_8(\epsilon)&=(\epsilon/\sigma)\exp[-2(\epsilon/\sigma)^2].
\end{aligned}
\tag{4}
$$

$b(\epsilon)$ is a box function which is equal to 1 in the interval $[-D:D]$ and zero otherwise. For $m=1,\ldots,6$ the ϵ integral thus extends from $-D$ to D, for $m=7,8$ it extends over the full real axis. For flat band ($m=1,2$) and the Bethe lattice ($m=3-7$), D has the meaning of the band half-width, for hypercubic lattice ($m=7,8$) it is just a rescale factor. By default, $D=1$ and

$\sigma = 1$, which are commonly used conventions. The expressions with the factor of ϵ are used in the expressions for the Hall transport[1].

$f(\omega) = 1/(1+e^{\omega/T})$ is the Fermi function at temperature T. We use the convention that the argument $\omega = 0$ corresponds to the chemical potential in G, ρ and f. The chemical potential μ only appears as the shift in the denominator of G.

n and o are some integers. $n = 2, 3$, $o = 0, 1, 2$ are of particular interest. For conductivity, $n = 2$, $o = 0$. For thermopower, $n = 2$, $o = 1$. For thermal transport, $n = 2$, $o = 2$. For Hall conductivity, $n = 3$, $o = 1$. For Hall thermoconductivity, $n = 3$, $o = 2$. (The code does not deal with the prefactors to I_{mno} to find these physical quantities, it just computes the dimensionless integral I_{mno} and it is up to the user to decide what to do with the resulting number. In particular, it is to be noted that ρ is defined here as the spectral function including the π factor.)

The input are the indexes m, n and o, temperature T, chemical potential μ, and tabulated $\Sigma(\omega)$, i.e., a file containing columns $\{\omega, \mathrm{Re}\Sigma(\omega), \mathrm{Im}\Sigma(\omega)\}$.

The output is I_{mno} (real quantity) and optionally an error estimate.

III. IMPLEMENTATION

The ϵ integrals

$$J_{mn}(\Omega) = \frac{1}{D}\int_{-\infty}^{\infty} \mathrm{d}\epsilon\, \Phi_m(\epsilon) \left[-\frac{1}{\pi}\mathrm{Im}\frac{D}{\Omega-\epsilon}\right]^n \tag{5}$$

are evaluated analytically in Mathematica for all m and $n = 2, 3$. These twelve (dimensionless) functions take Ω as input and return a real quantity as output. It may be assumed that $\mathrm{Im}\Omega > 0$ (and the code needs to assert that this is indeed the case). In practice, we will always take $\Omega = \omega - \Sigma(\omega)$ as argument.

The resulting expressions are to be converted to C code and saved to a file which will be included from the main program written in C or C++. This step of the code generation has to be fully automated (no copy/pasting!) and performed, for example, through evaluating a Mathematica notebook or script.

In the main program, $\Sigma(\omega)$ is read from a file and interpolated; the user can choose between linear and higher-order (cubic spline, Akima spline) interpolation. GNU scientific library (GSL) interpolation object are used for this purpose.

The ω integration

$$I_{mno} = \int_{-\infty}^{+\infty} \mathrm{d}\omega \left(-\frac{\partial f(\omega)}{\partial\omega}\right) J_{mn}[\omega - \Sigma(\omega)](\omega/D)^o \tag{6}$$

is then performed. The tabulated Σ defines the integral boundaries (ω_{min} and ω_{max}) to be used instead of $(-\infty, +\infty)$. In other words, $\mathrm{Im}\Sigma$ is assumed to be zero outside the interpolation interval.

The derivative of the Fermi function can be written as

$$-\frac{\partial f(\omega)}{\partial\omega} = \frac{1}{2T(1+\cosh x)} \tag{7}$$

where $x = \omega/T$. Because of this weight function, the main contribution to the integral comes from the interval $x \in [-5 : 5]$. The idea is thus to rewrite the integral I_{mn} in terms of the variable x,

$$I_{mno} = \int \mathrm{d}x\, g(x) J_{mn}[Tx - \Sigma(Tx)] \left(\frac{Tx}{D}\right)^o, \tag{8}$$

where $g(x) = 1/(2 + 2\cosh(x))$. The suitable upper limits to x integration have to be recalculated from ω_{min} and ω_{max}, but can be constrained within some sensible limits (e.g., -40 and 40) because $-\partial f/\partial\omega$ is exponentially decreasing and beyond 40 it is smaller than numerical round-off errors.

The integration is to be performed with the GSL quadrature routines with adaptive grid algoritms. It would be of interest to separately perform the integration for $x > 0$ and $x < 0$, with an option to display the partial results (if verbosity is increased using `-v`, see below).

IV. INVOCATION

The executable is named `bubble`. The syntax is:

```
bubble [-v] [-i <i>] [-p <p>] [-D <D>] [-s <sigma>] m n o T mu Sigmafile
```

The switch `-v` enables verbose output (e.g. error estimates). By default, only the final result I_{mno} is given as the output.

`i` controls the order of interpolation: $i = 1$ is linear interpolation, $i = 2$ is quadratic interpolation, $i = 3$ is Akima spline interpolation.

`p` controls the precision of the numerical integration.

The defaults are $D = 1$, $\sigma = 1$. They can be changed to `D` and `sigma`.

V. TESTING

The code can be tested using the limit of constant relaxation time, $\Sigma(\omega) = -i\Gamma$ with some constant $\Gamma > 0$. Then I_{mno} can be evaluated analytically for different T and the results compared to the output of `bubble`. A suitable input file then consists simply of two lines:

```
omega_min 0 -Gamma
omega_max 0 -Gamma
```

Testing also has to be automated and rerunnable (to be useful as a regression test).

Additionally, we want to test using real-life examples of $\Sigma(\omega)$ taken from DMFT(NRG) calculations for the Hubbard model, and compare against the known results for the transport quantities.

VI. CONCLUDING REMARKS

In the future we might be interested in additional functions $\Phi_m(\epsilon)$ corresponding to different lattices (e.g. square, cubic).

[1] L.-F. Arsenault and A.-M. S. Tremblay, "Transport functions for hypercubic and bethe lattices," Phys. Rev. B **88**, 205109 (2013).
[2] G. Bevilacqua, "Some integrals related to the fermi function," (2013).